

Manish Barti

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SUMMARY

Computer Science student with hands-on experience building production-ready web applications and backend systems using Python, FastAPI, PostgreSQL, and React. Proficient in REST API design, async programming, and cloud deployment (AWS, Docker). Seeking a software development internship.

EDUCATION

2023 – 2027 B.Tech, Computer Science & Engineering at **Graphic Era Deemed University**
Relevant: Data Structures & Algorithms, DBMS, Operating Systems, Machine Learning

2021 – 2023 Senior Secondary (CBSE), **Nirmala Convent Sr. Sec. School, Kathgodam**

EXPERIENCE

2024 – Present **Independent Project Developer** — Built and deployed full-stack and AI-powered systems independently;

PROJECTS

SafeNav – Web & Link Safety Analysis Platform | *FastAPI, PostgreSQL, ReactJS* [Demo](#) | [GitHub](#)

- Engineered a 3-tier heuristic engine (Lexical, SSL/TLS, IP Intelligence) with per-rule scoring, eliminating the black-box nature of traditional URL scanners.
- Minimized scan latency by executing network-bound threat intelligence queries concurrently via `asyncio.gather` in the FastAPI backend.
- Built an in-memory caching daemon syncing 100,000+ trusted domains via a background thread, optimizing API throughput and preventing false positives.
- Secured the backend with async PostgreSQL (SQLAlchemy) for scan history persistence and SSRF protections to block internal network scanning.

GigInsurance – AI Parametric Insurance | *LangGraph, LangChain, ChromaDB, XGBoost* [Demo](#) | [GitHub](#)

- Built a LangGraph autonomous claim agent classifying disruptions (rain, AQI, curfew) via Groq/LLaMA NLP across 6 real-time tools (Open-Meteo, government feeds, policy RAG) — decisions in under 30 seconds, zero human adjudication.
- Designed a hierarchical RAG pipeline with section-boundary-aware chunking (ChromaDB + `all-MiniLM-L6-v2`), co-locating payout thresholds and rules in atomic chunks to eliminate retrieval failures from fixed-character splitting.
- Architected a WhatsApp-first interface (Twilio / Meta Cloud API) on async FastAPI + `asyncpg` with OTP session auth and Alembic migrations for zero-install multilingual onboarding and claims over chat.
- Implemented a deterministic PHR × SLF payout engine (rs. 500/day cap) with XGBoost risk scoring and Isolation Forest anomaly detection to flag fraudulent claim clusters before payout.

SKILLS

Programming:	C++, Python, SQL
Web Development:	HTML, CSS, JavaScript, React.js
Backend & APIs:	FastAPI, Node.js, Express.js, REST APIs, <code>asyncpg</code> , Alembic
Databases:	PostgreSQL, MySQL, ChromaDB (Vector DB)
Generative AI:	LangChain, LangGraph, RAG, Hugging Face, FAISS, NLP
ML & Data:	XGBoost, Scikit-learn, Isolation Forest, Pandas, NumPy
Cloud & DevOps:	AWS EC2, AWS S3, Docker, Linux, Git
Tools:	GitHub, Postman, VS Code, PyCharm

CERTIFICATIONS

AWS Cloud Essentials	AWS Skill Builder	Jan – June 2026
AWS AI Practitioner	AWS Skill Builder	Jul – Nov 2025
Blockchain & Its Applications	NPTEL	Jan – Apr 2025